Developing Long Time Series 1-km Land Cover Maps From 5-km AVHRR Data Using a Super-Resolution Method

Haoyu Wang, Xiang Zhao, Shunlin Liang, *Fellow, IEEE*, Donghai Wu, Xin Zhang, Qian Wang, Jiacheng Zhao, Xiaozheng Du, and Qian Zhou

***Abstract*— Dynamic land cover (LC) information is an essential part of environmental and ecological research. Therefore, acquiring dynamic LC data with high spatial resolution has attracted a great deal of attention in the remote sensing community. Nevertheless, the high-temporal resolution satellite data tend to have a coarse spatial resolution, and satellite data with high temporal resolution are often relatively low. Obtaining LC with high spatiotemporal resolution is extremely challenging. The super-resolution method can help researchers achieve this goal, and the recently developed neural-network-based deep learning algorithms have great potential for use as an alternative solution. This study proposes a focal loss temporal convolutional long short-term memory (FL-T-ConvLSTM) model for super-resolution LC classification research. It first trains the deep FL-T-ConvLSTM network to establish a transformation between low-resolution quantitative remote sensing parameters and high-resolution quantitative remote sensing parameters and then engages in nonlinear mapping with a high-resolution LC map. A long-term series 1-km super-resolution LC classification model based on deep learning was established and applied to the Beijing–Tianjin–Hebei region. Based on this method, a long-term series of 1-km LC maps from 1982 to 2019 can be obtained. The test accuracy and field validation accuracy of the model reached 90.1% and 86.8% when using reliable test samples and field test samples, respectively. This study provides a method for obtaining high-resolution LC classification products from low-resolution quantitative remote-sensing products.**

***Index Terms*— Focal loss temporal convolutional long short-term memory (FL-T-ConvLSTM), land cover (LC), long time series, quantitative remote sensing parameters, super-resolution.**

I. Introduction

AS THE most prominent feature of the surface, land cover (LC) has received increasing attention [1], and it is a common result of human activities and natural evolution [2]. It is the key to understanding the interrelationship between human activities and natural ecological changes, and it has become an important factor that affects economic development and environmental issues [3]. Affected by social development, environmental change, and various factors, LC change has become a key component of global change research [4]–[6]. Due to applications of LC data, dynamic LC change information is being used in many current models [7], [8] and global change researches [9]–[12].

The continuous development of remote sensing technology has resulted in a variety of global or regional LC products. The existing LC classification products can be roughly divided into three categories according to the level of spatial resolution. The first category is the high-resolution classification products with a spatial resolution of 10–30 m. Among these products are Finer Resolution Observation and Monitoring of Global Land Cover (FROM-GLC) [13], FROM-GLC10 [14], Global 30-meter Fine Land Cover [15], and GlobeLand30 [16]. The second category is the medium-resolution classification products with a spatial resolution of 300–1000 m. The more representative datasets are MODIS LC classification data (2000–2018) [17], ESA's CCI dataset (1992–2018) [18], and China's multiperiod land use LC remote sensing monitoring dataset (CNLUCC) [19]. The third category is a coarse-resolution classification product with a spatial resolution of more than 1000 m. The existing products include the NOAA/NASA Pathfinder Advanced Very High Resolution Radiometer (AVHRR) Land Data Set (1982–2000) [20] and Global Inventory Modeling and Mapping Studies (GIMMS) land use and LC LULC maps [21]. The spatial resolution of these two datasets is 8 km. This category also includes two LC classification datasets with a spatial resolution of 5 km, the GLASS-GLC [22] and MODIS 5-km LC datasets [17].

Although many LC classification products have been produced, these data still have some limitations. For classification

Manuscript received April 22, 2020; revised July 27, 2020; accepted August 2, 2020. Date of publication September 3, 2020; date of current version June 24, 2021. This work was supported by the National Key Research and Development Program of China under Grant 2016YFA0600103 and Grant 2016YFB0501404. *(Corresponding author: Xiang Zhao.)*

Haoyu Wang, Xiang Zhao, Qian Wang, Jiacheng Zhao, Xiaozheng Du, and Qian Zhou are with the State Key Laboratory of Remote Sensing Science, Beijing Normal University, Beijing 100875, China, also with the Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences, Beijing 100875, China, and also with the Beijing Engineering Research Center for Global Land Remote Sensing Products, Faculty of Geographical Science, Institute of Remote Sensing Science and Engineering, Beijing Normal University, Beijing 100875, China (e-mail: why0925@mail.bnu.edu.cn; zhaoxiang@bnu.edu.cn; qianwang@bnu.edu.cn; zhaojiacheng@mail.bnu.edu.cn; duxzheng@mail.bnu.edu.cn; qian.zhou@mail.bnu.edu.cn).

Shunlin Liang is with the Department of Geographical Sciences, University of Maryland, College Park, MD 20742 USA (e-mail: sliang@umd.edu).

Donghai Wu is with the Department of Ecology and Evolutionary Biology, Cornell University, Ithaca, NY 14853 USA (e-mail: dw623@cornell.edu).

Xin Zhang is with the State Key Laboratory of Remote Sensing Science, Institute of Remote Sensing and Digital Earth, Chinese Academy of Science, Beijing 100101, China (e-mail: zhangxin1010@radi.ac.cn).

Digital Object Identifier 10.1109/TGRS.2020.3018109

1558-0644 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.

data with high spatial resolution, retrieving annual LC products is very difficult because of the large amount of data and uneven quality [23]. The existing medium-resolution LC products can obtain annual classification data. However, the earliest starting point for these products is 1992, and there was a lack of LC products before the 1990s. Although the availability of coarse-resolution LC products began in 1982, their spatial resolution is relatively coarse [8]. Due to the lack of LC products with high spatiotemporal resolution, the alternative method assumes that the LC is unchanged [4], [24]–[27], or it uses the existing classification data to extract invariant pixels to obtain stable LC [28]. Because human activities and ecological changes are dynamic, LC is also constantly changing [29]. The existing alternatives can cause uncertainty in the results of climate change models or surface change studies.

Long-term serial LC classification products have great application potential [30]. However, we do not know how to obtain long time series LC products. The primary problem at present is that the existing data cannot have both high temporal resolution and high spatial resolution, and there is a particular lack of continuous LC data before 2000. AVHRR data have the longest time span among remote sensing data [31], but their spatial resolution is relatively coarse. To solve these problems, the most feasible method at present is to use a super-resolution method to extract high-resolution LC from low-resolution data. Combined with the high temporal resolution of AVHRR data, it is possible to obtain high spatial resolution long time series LC classification data. Therefore, this study was intended to obtain a set of 1-km LC classification products from 1982 based on the low-resolution long time series AVHRR data through the super-resolution method.

Using a super-resolution method based on deep learning is the most effective method of obtaining LC classification data with high spatiotemporal resolution from low-resolution images [32]–[37], and the central idea of these methods is to input low-resolution images and establish nonlinear transformation between low-resolution data and high-resolution data through deep learning models [35], [38]. The super-resolution method primarily extracts subcategories by subpixel decomposition based on the spatial dependence between the different types of LC [39]. A common algorithm is used to reconstruct the spatial distribution and spectral features of images through deep learning, for example, SRCNN [40], FSRCNN [32], and ESPCN [41].

Although there are multiple deep learning super-resolution methods, these super-resolution mapping methods perform super-resolution image restoration through spatial dependence between pixels [42]. There are many time series data with rich information, but it is difficult to use it directly in a super-resolution model [44]. This limitation occurs mainly because time series data are very complex, and too much data can lead to information redundancy. Therefore, there is a need for an efficient way to use dynamic quantitative remote sensing products [45] for classification. The LSTM model [46] can capture time correlation and mine time information from data [47], [48]. The convolutional neural network (CNN) model [49] can extract local features [50]–[53] and perform dimensionality reduction to reduce the computational complexity by pooling. As a result, the characteristics of these two models can be very helpful for extracting time series information. There is currently an LSTM model that can perform convolution work called ConvLSTM [54]. The ConvLSTM model is suitable for processing time series data with rich spatial information. These types of data are rich in spatial information and have strong spatial correlations. Therefore, these data are generally 3-D and have large spatial range. The samples for this study were derived from the CNLUCC data. Considering the accuracy of this product, we used sliding windows to select highly reliable samples. These samples are not continuous in space after selection. Neither the data form nor the amount of data is applicable to ConvLSTM. The use of ConvLSTM with spatial convolution may affect the model performance.

Therefore, this study proposes a deep learning method called the focal loss temporal convolutional long short-term memory (FL-T-ConvLSTM) network. This model was intended to use the CNN for long-term serial remote sensing data smoothing through a rectangular convolution kernel. Then, the processed data were input into the LSTM model to extract dynamic information. The surface quantitative remote sensing parameters were extracted by the model and were then input into the softmax layer for multiclass classification to obtain the final super-resolution classification result. Based on this method, annual LC data with a 1-km spatial resolution can be obtained. The method has high computational efficiency, and it can meet the demands of practical applications over a large area.

II. Super-Resolution Land Cover Classification Based on FL-T-ConvLSTM

The framework of the 1-km super-resolution LC classification is shown in Fig. 1. The entire process is based on the TensorFlow's deep learning framework. TensorFlow is an open-source machine learning platform that uses data flow graphs. It is widely used in the programming of various machine learning algorithms. Because of its comprehensive and flexible nature, it allows developers to easily build applications supported by machine learning.

A. T-ConvLSTM Combining the CNN and LSTM Network

Time series data represent a series of data obtained in chronological order that can reflect the temporal variation in the study object. The LSTM networks are commonly used to handle sequential information in various problem domains, such as natural language processing and text or speech generation. Compared with a single-time model, the LSTM can theoretically process unlimited amounts of data and make decisions during the actual time series [55], [56]. Therefore, LSTM is ideal for processing time series data.

The CNN is a multilayer feedforward neural network. Its 2-D structure-oriented convolution pooling method helps investigators grasp the essential features of images for classification. Convolution and pooling operations are the core parts of CNN [57]. Through the convolution operation, we can extract the features of the image, and the convolution operation can enhance certain features of the original signal and reduce

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.

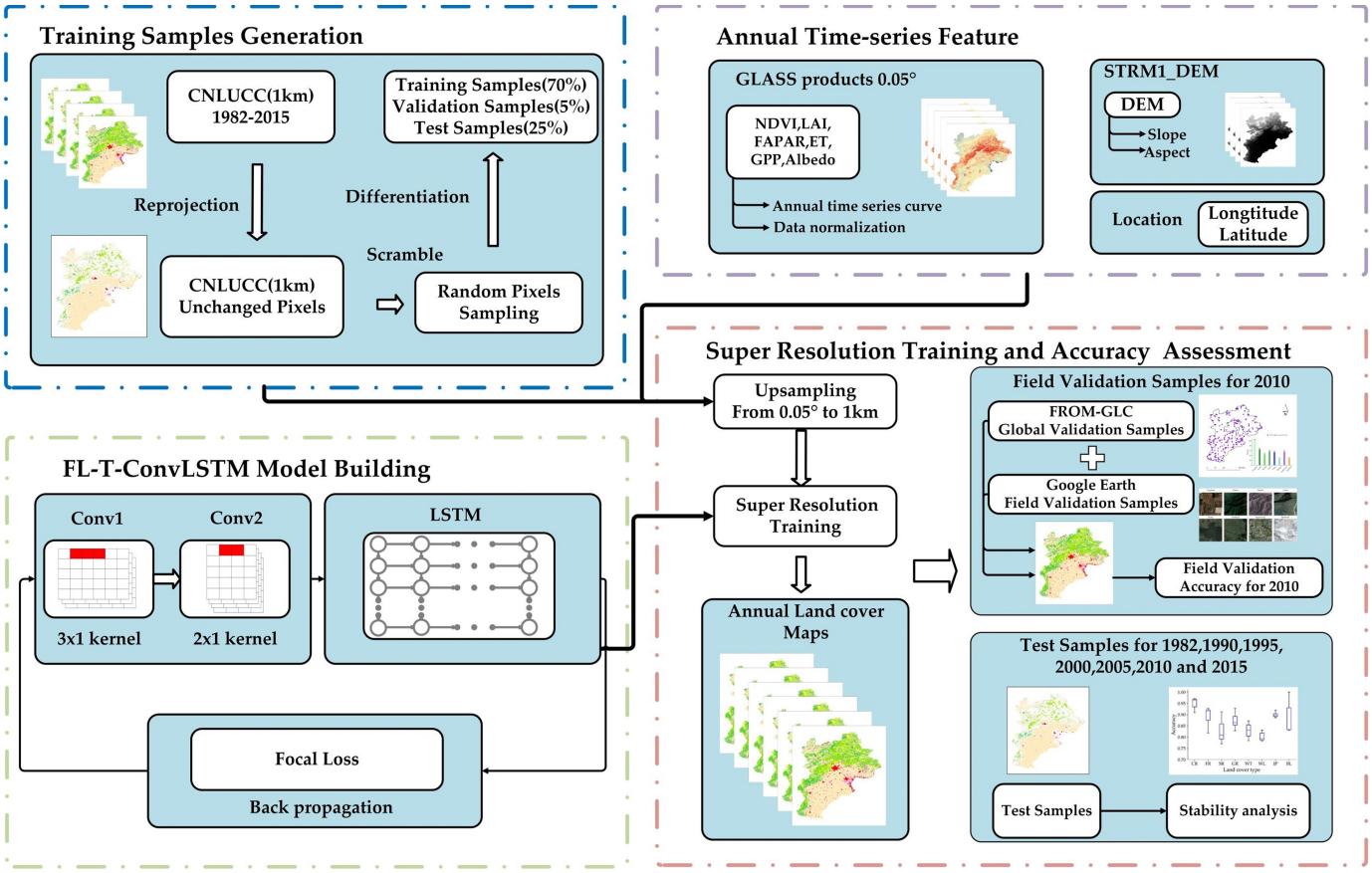


Fig. 1. Framework for building the annual dynamics of LC classification with 1-km superresolution, including the preprocessing of classified data and quantitative remote sensing product data, model building, and the evaluation of the results.

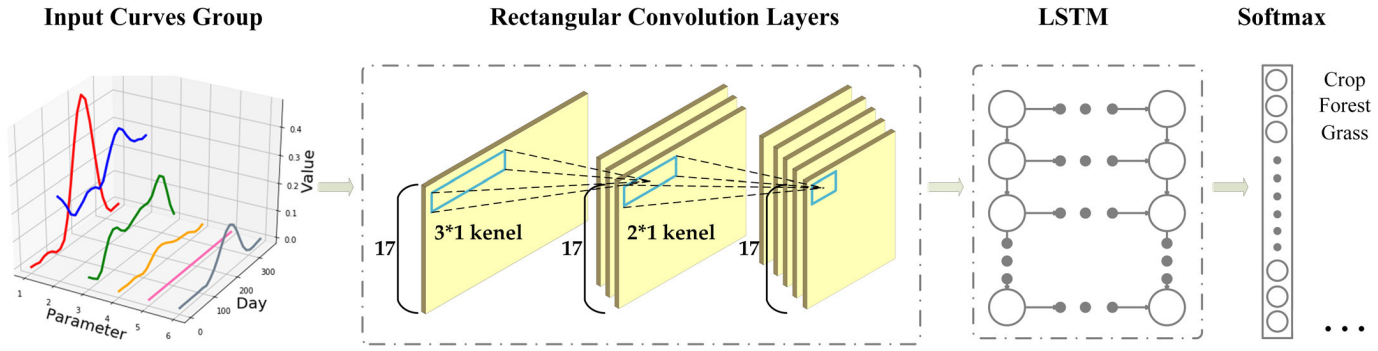


Fig. 2. T-ConvLSTM model. The first two layers are CNNs using rectangular convolution kernels, and then, they are connected to the LSTM network. Finally, the softmax layer is used for classification. We input a total of 17 surface features. Therefore, the width of the matrix during convolution is 17 and does not change.

the noise. After the convolutional work, the model downsamples the image, which reduces the amount of data processing while preserving useful information.

To use LSTM to process time series data and combine the advantages of the CNN extraction features, this study proposes an LSTM network that combines the CNN feature extraction capabilities called—temporal-ConvLSTM (T-ConvLSTM) (Fig. 2). This network advanced the convolution operation before entering the time series data into the LSTM model. However, the convolution method is different from the commonly used CNN. The commonly used CNN convolution kernel is a square structure. The advantage is that feature extraction of the spatial information from the image can be performed. However, for time series parameters, the expression of the data is time curve data that extend from the time axis. Therefore, the convolution of a square is not suitable for processing these data. The sample feature set in this study is a collection of time series information for various quantitative remote sensing products. The time series characteristics of each quantitative remote sensing session are as follows: $(h^1, h^2, \ldots, h^{T-1}, h^T)$. Combined with the latitude and longitude elevation data, each sample feature is combined

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.

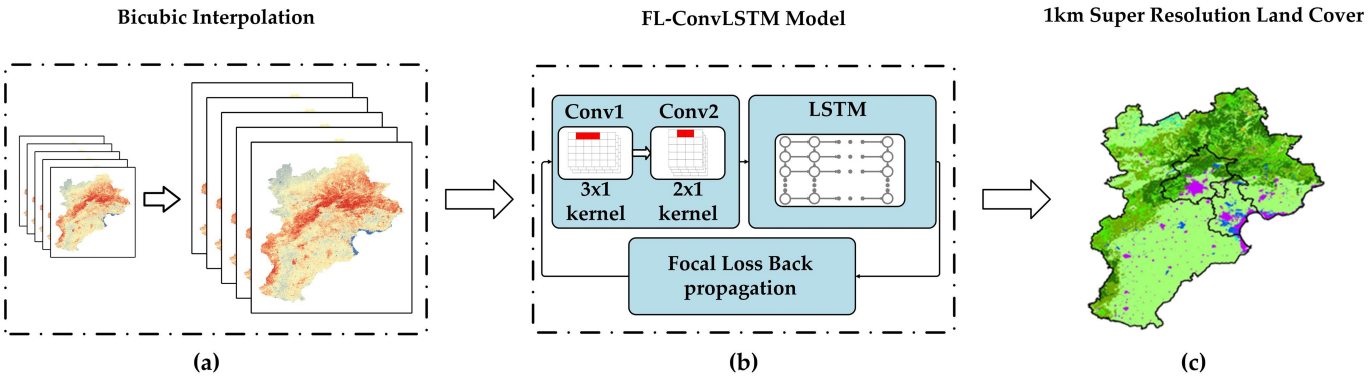


Fig. 3. Superresolution classification model. (a) Interpolation work before the image input model. (b) Primary model used in the study. (c) Result of the model prediction.

into a 46 × 17 matrix. For this matrix, the general CNN will use a square convolution to convolve the 46 × 17 matrix. The axis of length 17 is set as the Y axis, and the axis of length 46 is set as the X-axis. The Y-axis values represent different quantitative remote sensing features, and performing convolution on this axis does not make sense. The convolution kernel used in this study is an $X \times 1$ rectangular convolution kernel. The convolution kernel has a length of 1 between different quantitative remote sensing features and length X in the time series. Therefore, the convolution kernel will only slide and convolute in the time series direction.

B. Focal Loss for Imbalanced Samples

Having an imbalanced sample size is a very common and difficult problem to solve in machine learning classification [58], [59]. Since the model/algorithm has never adequately trained all the category information, it will result in an unbalanced category that cannot be optimized. A small number of observations of a small sample class will create a problem with a validation or test sample, which is difficult to represent in a class. If the sample of a category has a significant advantage over other classes, the dataset can be considered an unbalanced dataset. This imbalance will lead to two problems: 1) training is inefficient because sample imbalances can cause insufficient training in some categories and 2) due to insufficient training in some categories, the performance of the model will decrease.

There are currently two primary approaches to this problem. Undersampling [60] eliminates the number of classes in the data by randomly deleting classes with enough samples. Oversampling [61] increases the number of observed samples randomly for unbalanced classes (classes with a small number of samples). Both the methods address the sample set. Although the sample size is balanced, it could have an impact on the representativeness of the sample. Therefore, this study is expected to use the model to learn the characteristic category information fully from the limited samples using the model; in other words, to solve the sample imbalance problem through the balanced training of the model.

When sample imbalance occurs, the category with a large number of samples will play a major role in the loss, which will dominate the direction of the gradient update. This deep learning network will not be able to adequately learn useful information in a category with a small number of samples, which would result in a large number of misclassifications in categories with small sample sizes. Focal loss [60] is used to solve the category imbalance problem by changing the training weights of the number of unbalanced samples. Thus, even if the number of samples in a particular category is large, they contribute little to the total loss. This method can focus the model on the class with a smaller sample number, and it would increase the model's attention to this category to solve the imbalance learning phenomenon in the model.

The focal loss for multiclassification [62] can be calculated simply using the following formula:

$$\mathrm{FL}(p_l) = -\alpha \times (1 - p_l)^{\gamma} \times \log(p_l) \tag{1}$$

where α is the class weight, γ is the focusing parameter, and p_l is the value of the cross-entropy of category l.

C. Superresolution LC Classification Model

This study establishes an end-to-end mapping of 0.05° quantitative remote sensing products to 1-km LC classification. The map is represented as T-ConvLSTM using focal loss (FL-T-ConvLSTM), which used a low-resolution 0.05° quantitative remote sensing product as the input and outputs a 1-km LC classification image (Fig. 3).

Superresolution is characterized by the ability to obtain fine images based on rough data. The current research is primarily based on the conversion of the image detail and resolution on the same image. Deep learning for super-resolution image conversion is usually a resolution conversion based on continuous-valued images, which is learned by a deep learning model. The deep learning LC classification links the surface feature parameters with the LC categories and performs LC classification work. Both these tasks are based on a deep learning model that establishes the end-to-end relationship.

Therefore, this study was intended to connect these two processes in series. In the first step, continuous low-resolution quantitative remote sensing features are converted into high-resolution features. After that, high-resolution features are correlated with classification labels to form a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.

TABLE I
1982, 1990, 1995, 2000, 2005, 2010, AND 2015 UNCHANGED SAMPLE SET FOR MANY YEARS. CROP (CR), FOREST (FR), SHRUBS (SR), GRASS (GR), WATER (WT), GLACIER (GL), WETLAND (WL), IMPERVIOUS (IP), AND BARE LAND (BL)

	1982-2000	2000-2015	Training	Validation	Test
CR	320211	419172	70%	5%	25%
FR	45321	72356	70%	5%	25%
SR	15729	28564	70%	5%	25%
GR	36223	56636	70%	5%	25%
WT	4863	4992	70%	5%	25%
WL	120	996	70%	5%	25%
IP	6378	12056	70%	5%	25%
BL	171	228	70%	5%	25%

super-resolution classification model. Since both the processes can be expressed during deep learning, we use low-resolution features directly as input and then use high-resolution LC data as the output. This study used only one model to try to establish a mapping relationship between two factors.

The specific approach is as follows. First, the low-resolution 0.05° quantitative remote sensing product is amplified to the desired size by bicubic convolution interpolation, which is represented as Y. It has the same size as the high-resolution 1-km LC classification, which is represented as X. The amplified high-dimensional vector is then extracted from it as an input to the model. The second step is to establish a nonlinear mapping. This operation converts each high-dimensional variable (Y) to another high-dimensional variable (X). The model learns the mapping between X and Y through continuous training. The final step is the superresolution reconstruction work. The process is to convert low-resolution images through the model to generate the final high-resolution LC classification.

Unlike traditional superresolution reconstruction, the interesting work of superresolution classification is that the average block extraction and overlapping high-resolution regions can be eliminated. Superresolution classification can directly connect Y with a single-pixel in X to achieve unit-to-unit docking. This approach will simplify the workload of the super-resolution classification.

III. SAMPLE SET ACQUISITION AND FEATURE COLLECTION

A. Training Sample Set

The sample set was collected from the CNLUCC [19] dataset.

Due to the accuracy of the dataset itself, there is uncertainty associated with selecting samples from it directly. If a land type is unchanged for many years, the land type of the area is relatively stable. Therefore, the accuracy and purity of the samples obtained from this area are relatively reliable (Table I).

Two sets of invariant pixel samples were screened out. The first is the invariant pixels from 2000, 2005, 2010, and 2015. This sample set is used for comparison between the superresolution method and the direct inversion method. The second sample set was composed of the three-year invariant pixels in 1982, 1990, and 1995 (Table I). This sample set is used for experimentation on and validation of the superresolution method for LC classification before 2000.

B. Field Validation Sample Set

The validation of the classification results is related to the final result of the entire LC classification study [63]. If the classification results are not properly verified, then any LC mapping is an untested hypothesis and cannot be used without any doubts. Therefore, this step is critical to the selection and use of validation data [64], [65]. The sample set in this study was obtained based on the CNLUCC data. Although it has undergone a series of screenings, its misclassification does not fully guarantee the accuracy of the sample. In addition, it cannot remove the suspicion of "self-validation." Therefore, obtaining a set of third-party datasets and using them for validation is very important.

Tsinghua University's Global Land Cover Validation Point Dataset (2010) is a sample of the global LC data production process for FROM-GLC [13]. This study extracted the Beijing–Tianjin–Hebei area of the dataset for the final validation of the classification results. However, after vector extraction, there were only 55 validation sample points for Beijing–Tianjin–Hebei, and there were only six categories with the distribution. Among these categories, there are 16 sample sites in the cropland category, whereas the other categories have less than 10. Validation with so few sample points can lead to contingent results. Therefore, more field validation samples are needed.

This study draws on the extraction method of the global LC validation point dataset. A set of 2010 field validation point datasets were screened for each category on Google Earth by visual interpretation and expert knowledge [66]–[68]. During the screening process, we attempted to ensure that the spatial resolution of Google Earth was approximately 1 km. Under the condition that the spatial distribution is roughly uniform, the spatial detail feature (graph) is viewed by zooming in to determine the category attribution of the point (Fig. 4). We obtained a set of field verification samples based on this method.

After the validation sample point extraction is completed, the investigator compares the data with the CNLUCC and MODIS 1-km LC datasets, and it hands the dataset to other researchers for review. After correction, a set of 2010 LC field validation point datasets for the Beijing–Tianjin–Hebei region were formed. During the sample screening and extraction process, this study is intended to ensure that the sample points can be evenly distributed throughout the study area, and it generally follows the principle of stratified sampling. Finally, the acquisition of the field validation sample point set was completed. The spatial distribution and quantity distribution of the sample point set are shown in Fig. 5.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.

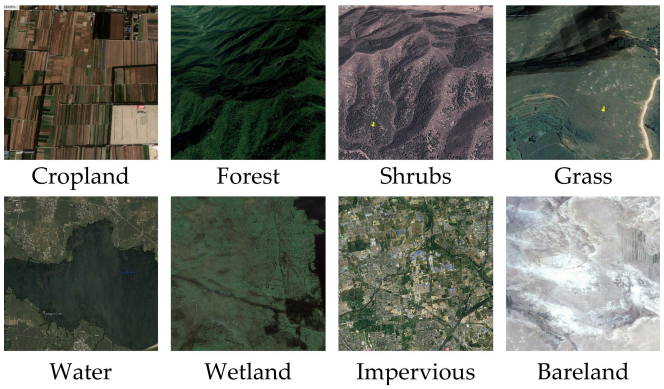


Fig. 4. Representative exemplars of each LC type from Google Earth.

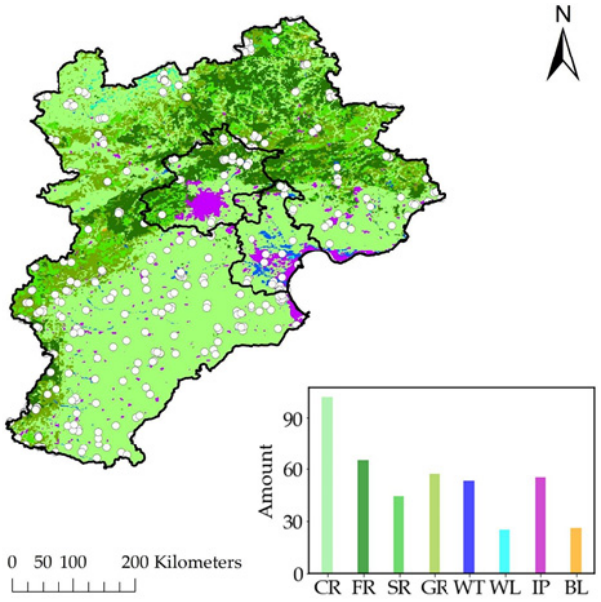


Fig. 5. Validation samples' distribution. This figure shows the distribution of field verification points and the number of field verification points in different categories. The Y-axis of the histogram is the number of statistics, and the X-axis shows the category of the verification point.

C. Feature Collection for the Classified Sample Set

The input data for this study were the GLASS data (Liang *et al.* [43]), including the leaf area index (LAI) [69], evapotranspiration index (ET) [70], fractional absorption of photosynthesis active radiation (FAPAR) [71], gross primary productivity (GPP) [43], and Albedo [72]. We used quantitative remote sensing products at two resolutions (1 km and 0.05°).

The 0.05° product was used for super-resolution studies, and the 1-km product was used for comparative testing.

The surface characteristics of different features are different, which is the basis for LC classification. For a certain type of land, its surface characteristics also have their own unique rules of change during the year. Based on this circumstance, this study constructed a feature set that includes changes in the year, phrenological characteristics, topography, and spectral index. Here, a variety of surface features were extracted by the GLASS quantitative remote sensing products based on AVHRR. Due to the systematic deviation of the AVHRR data itself, this study performed time series feature reconstruction on the yearly time series data, to reduce the time series noise. We used the Savitzky–Golay (SG) filtering method [73]. This filtering method can filter out noise while keeping the shape of the curve and retaining the curve provides as much detail as possible. After the correction, the fluctuation of the features becomes small, and the respective outliers are also adjusted (Fig. 6). This study also calculated the slope and aspect information from the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) DEM data and combined the latitude and longitude as part of the feature set. Finally, the annual time series feature set was constructed from 1982 to 2018, including annual GLASS time series feature information, NDVI annual time series information, auxiliary DEM, slope and aspect information, and latitude and longitude information.

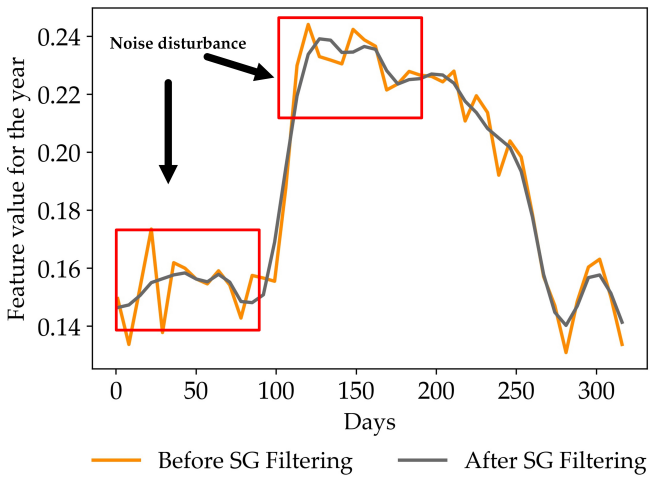


Fig. 6. Characteristic curve after SG filter reconstruction.

IV. Experimental Results and Discussion

A. Comparison of Different Classification Models

The performance of several different machine learning models (Random Forest [74], SVM [75], SRCNN, LSTM [56], T-ConvLSTM, and FL-T-ConvLSTM) in super-resolution LC classification was compared (Table II). The difference between T-ConvLSTM and FL-T-ConvLSTM is in whether or not the focal loss is used for weighting during training. In addition, we also used the classic SRCNN model for experiments. The commonly used super-resolution deep learning models include SRCNN, FSRCNN, ESPCN, and so on[76]. The effects of these models are roughly the same, but the difference is only in the convolution approach. The first step of FSRCNN and ESPCN is to perform convolution on the spatial dimension. Since the sample set in this study was selected randomly with high accuracy, the sample set has lost spatial continuity and correlation. We cannot perform spatial convolution using ESRCNN and ESPCN on the training samples in this study. Therefore, they are not suitable for the classification work in this study. Considering the characteristics of our data, SRCNN was chosen as the comparison model. Unlike the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON OF SIX MODELS

Method	Accuracy	Mean f1-score
SVM SR	75.3%	0.751
SRCNN	77.7%	0.773
LSTM SR	86.3%	0.859
Random Forest SR	86.7%	0.864
T-ConvLSTM SR	89.1%	0.889
FL- T-ConvLSTM SR	90.1%	0.901
FL- T-ConvLSTM	91.2%	0.913

general SRCNN, the CNN model in the SRCNN method of this study is used to convolve the time series feature matrix of each sample.

Simultaneously, we also used the 1-km resolution quantitative remote sensing product as input to directly train the 1-km LC classification model as a comparison standard for super-resolution effects. This study tested the performance of seven models in super-resolution LC classification. The accuracy of the seven models reached 75.3%, 77.7%, 86.3%, 86.7%, 89.1%, 90.1%, and 91.2%. In the superresolution classification method, the FL-T-ConvLSTM model performed the best. This finding demonstrates the advantages of T-ConvLSTM in quantifying products in time series and extracting time series feature information.

In addition, this study also compared the super-resolution classification method with the direct classification method. The direct method inputs 1 km of the GLASS products into the model. The overall accuracy of the direct method was found to be 90%.

The highest precision of this superresolution classification method can reach 89.7%. This performance is almost in direct proximity to the direct classification method. It proves the practical feasibility of the super-resolution classification method from 5 to 1 km.

B. Accuracy Validation of the Superresolution Classification Model

The test accuracy of the superresolution model is 90.1% (Table III). The classification result confusion matrix is shown in Table III. There are five land types with an accuracy of over 80%., and the accuracy of the cropland, forest, grassland, water, and impervious analysis reached more than 80%. The type of wetland is relatively low, primarily because of the indivisibility of the wetland category itself. The shrub category is similar to the forestland and grassland, and it has similarities in its surface features; thus, it is also difficult to distinguish. Since the quantitative remote sensing features are obtained by the super-resolution method, the edge features of each category are blurred due to the superresolution ambiguity of each category, which can easily cause confusion.

TABLE III
CONFUSION MATRIX BASED ON THE TEST SAMPLES

	CL	FR	SR	GR	WT	WL	IP	BL	Total
CL	176590	1443	1082	4344	439	141	771	35	95.5%
FR	371	26395	1408	1193	26	0	26	0	89.7%
SR	199	1040	8707	1115	6	0	0	6	78.6%
GR	1258	573	817	20481	26	40	7	13	88.2%
WT	355	32	8	8	2006	0	39	16	81.4%
WL	66	0	4	9	0	199	0	0	71.4%
IP	578	0	6	18	12	0	3994	0	86.7%
BL	13	0	0	13	0	0	0	73	73.3%
Total									90.1%

TABLE IV
CONFUSION MATRIX BASED ON FIELD SAMPLES

	CL	FR	SR	GR	WT	WL	IP	BL	Total
CL	97	1	0	1	0	0	3	0	95.10%
FR	1	49	1	4	1	0	0	0	87.50%
SR	0	1	41	0	0	0	2	0	93.18%
GR	2	5	2	45	0	1	0	2	78.95%
WT	2	2	1	0	45	2	0	1	84.91%
WL	1	1	0	2	0	17	2	0	73.91%
IP	6	1	0	0	1	0	47	0	85.45%
BL	1	2	0	3	0	0	0	20	76.92%
Total									86.78%

The test samples were extracted from the CNLUCC data. Although the data have been filtered by reliability pixels, it is highly desirable to select third-party validation data for accuracy testing because of the uncertainty of the CNLUCC data. This study screened a field validation sample set through Google Earth. We extracted the model output of the geographical location of the field validation sample and used it to test the accuracy (Table IV). The overall accuracy is 86.78%, with the accuracy of the cropland and shrubs being above 90%, and the accuracy of the grassland, wetlands, and bare land below being 80%. The primary reason why the grassland accuracy is less than 80% is that five pixels are misclassified as woodland. This finding primarily occurred because both are vegetation and are easily confused. Part of the reason why wetlands and bare land are less than 80% is that the wetlands by themselves are not easy to distinguish, and the other reason is because the two categories each have a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.

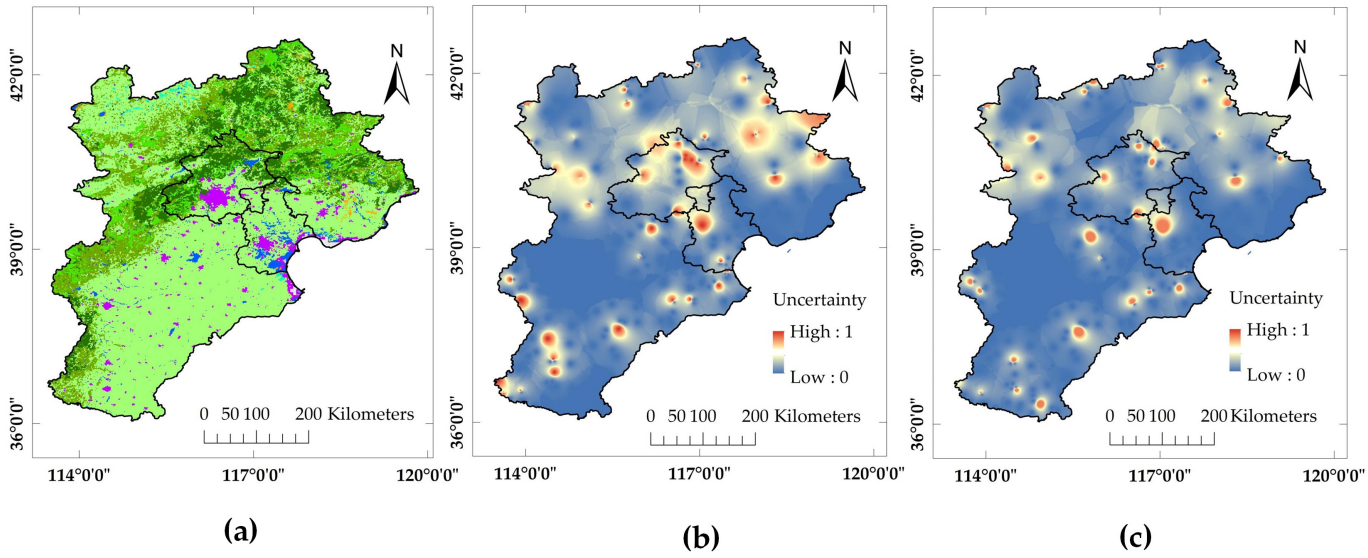


Fig. 7. Uncertain geographic distribution of 1-km super-resolution LC in the Beijing–Tianjin–Hebei area in 2015. The red areas indicate higher uncertainty value, and the green areas show lower uncertainty value. (a) Classification results in 2015. (b) Uncertainty testing for superresolution classification results based on the 5-km GLASS quantitative remote sensing products in 2015. (c) Uncertainty testing of the classification results based on the 1-km GLASS quantitative remote sensing products in 2015.

small sample size and are prone to uncertainty. Although the number of field samples is small, there is a certain uncertainty in the validation results. However, the accuracy of 86.78% can be accepted for the current classification study.

Since the confusion matrix cannot intuitively uncover the advantages and disadvantages of the classification results from the spatial distribution, this study further explores the uncertainty in the classification results to discuss areas with difficult classification from a spatial perspective. The correctly classified positions in the field-verified samples were scaled as 1, and the incorrect positions were scaled as 0. Thus, the spatial distribution of uncertainty was drawn [77]. The uncertainty value is between 0 and 1. A smaller value indicates lower uncertainty in the area. The closer the value is to 1, the more likely the regional category will be misclassified.

The classification results from the superresolution 1-km super-resolution classification and the field verification sample points were put together to generate a spatial distribution of the LC uncertainty. As shown in Fig. 7, most areas in Beijing, Tianjin, and Hebei have low uncertainty values, which indicates that the 2015 results are highly credible. Some areas are still yellow or orange, whereas a few areas are red, which indicates areas that could be misclassified. The areas with high uncertainty are in the forest, grassland, and shrub interlaced areas in the western parts of Beijing–Tianjin–Hebei. We found that the uncertainty of the vegetation coverage area in northern Beijing is relatively high; the primary reason could be that the vegetation distribution in the area is too complex to distinguish.

C. Model Stability Analysis

The primary purpose of this study is to develop a long-term series for a 1-km LC dataset. The long-term series LC data are based on the long-term series quantitative remote sensing products. The previous work proved that the super-resolution method is feasible for use in a 0.05° quantitative remote sensing product after 2000. To prove the versatility of the method, this study also trained a super-resolution classification model using quantitative remote sensing products from the year 2000. When comparing the test results of the model with the results after 2000, it is clear that the method is also applicable to quantitative remote sensing products before 2000 (Fig. 8). The accuracy gap between each category before 2000 and after 2000 is no more than 7%. The average error is 2.8%, and the maximum error is 6.1%. The model has roughly the same accuracy for each category before 2000 and after 2000. This finding proves the feasibility of the method for long-term series1-km LC classification.

This study has high requirements for its time generalization ability. Long-term series LC classification data have high requirements for the consistency of classification accuracy. Therefore, the generalization ability of the model needs to be highly valued. The super-resolution classification model was tested separately from the existing seven-year (1980, 1990, 1995, 2000, 2005, 2010, and 2015) validation sample set. This study counts the standard deviation of the accuracy over the seven years. It was found that the error of each type was not greater than 5.1%, and the overall accuracy error remained at 0.7%. It turns out that the model provides good time generalization.

D. Comparison of the Superresolution Method and Direct Inversion Method

LC classification by a super-resolution algorithm has yielded good results. However, the similarities and differences between the super-resolution method and the direct inversion algorithm [45], [78], [79] by the 1-km quantitative remote sensing products (no super-resolution method) are worth

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.

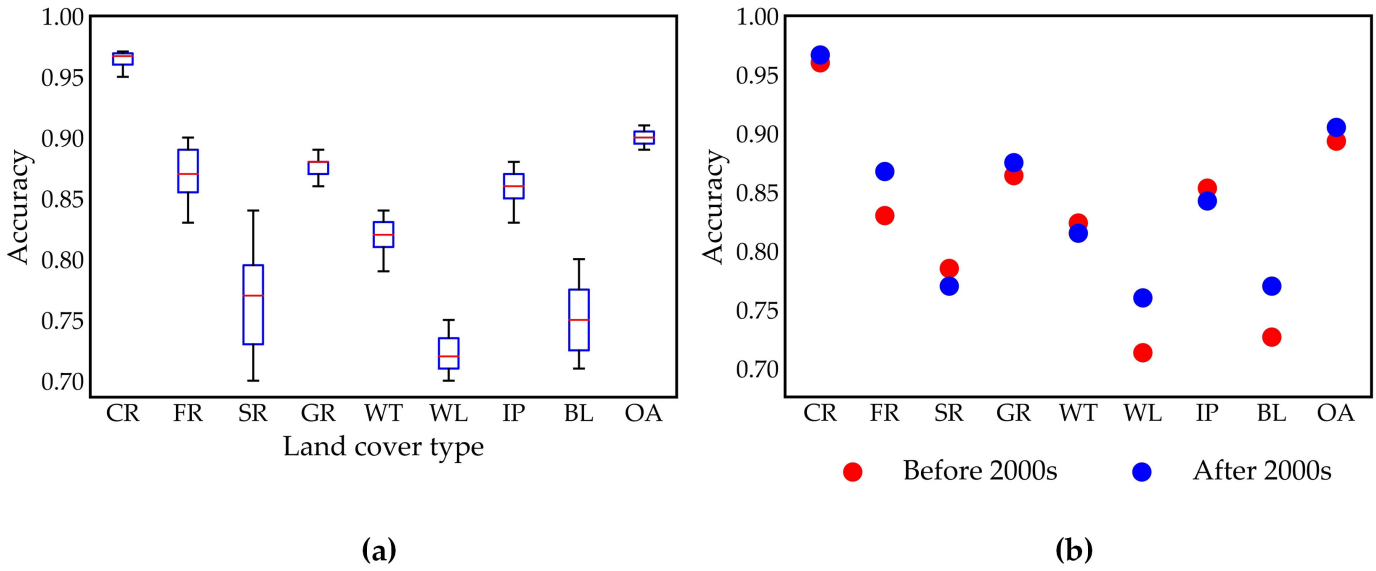
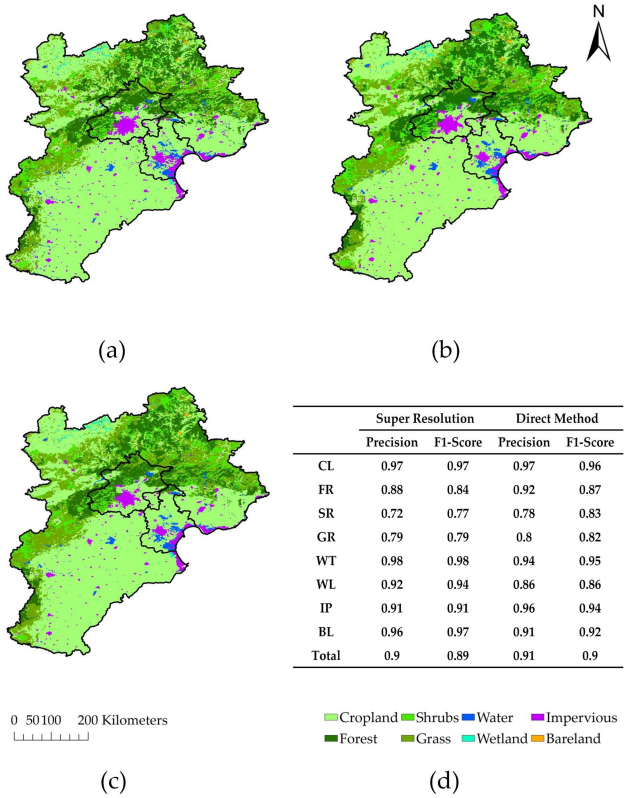


Fig. 8. (a) Box diagram of the accuracy of the test set for different years. (b) Accuracy comparison of the super-resolution models before the 2000s (red dots) and after the 2000s (blue dots). The Y-axis of the histogram is the classification accuracy, and the X-axis is the category of the verification point.

additional discussion. Therefore, this study compared the super-resolution method with the direct inversion algorithm. To quantify the difference between the two methods, this study trained two sets of LC classification models. The first uses the super-resolution method to train an FL-T-ConvLSTM super-resolution classification model; the second trains the FL-T-ConvLSTM classification model directly with the 1-km GLASS parameters as input.

The test accuracy of the two models reached 91.2% and 90.3%, respectively (Fig. 9). The accuracy of the direct inversion algorithm is slightly higher, primarily because the details on the surface features of the 1-km quantitative remote sensing parameters are more obvious. However, the accuracy of the super-resolution method is almost comparable to the traditional methods. The gap between the types of impervious layers is more obvious, with a difference of 5%. The primary misclassification is that the type of impervious layer is misclassified into cropland. Since the impervious layer is mainly urban cover, there will be a large number of interlaced areas of artificial ground and cropland on the edge of the city. Therefore, this circumstance can easily cause a misclassification phenomenon [80]. During the image restoration process, the super-resolution method is always prone to blurring. Therefore, for the boundary between urban areas and cropland, the super-resolution method is more prone to misclassification. However, this misclassification phenomenon is rare and is currently within an acceptable range.

For super-resolution boundary problems, a similar phenomenon occurs in each category [81]. The misclassification phenomenon also occurs primarily in the edge interaction zone of each category. The primary problem of misclassification is that other categories are misclassified into the cropland category. The primary reason is that because of the high sample size of the cropland, most of the information learned by the model is characteristic of the cropland. In addition,



	Super Resolution		Direct Method	
	Precision	F1-Score	Precision	F1-Score
CL	0.97	0.97	0.97	0.96
FR	0.88	0.84	0.92	0.87
SR	0.72	0.77	0.78	0.83
GR	0.79	0.79	0.8	0.82
WT	0.98	0.98	0.94	0.95
WL	0.92	0.94	0.86	0.86
IP	0.91	0.91	0.96	0.94
BL	0.96	0.97	0.91	0.92
Total	0.9	0.89	0.91	0.9

Fig. 9. (a) Comparison of the LC classification data. CNLUCC data. (b) Superresolution algorithm by the 5-km quantitative remote sensing products. (c) Direct inversion by the 1-km quantitative remote sensing products. (d) Comparison of the accuracy of the superresolution method and direct inversion method in different categories.

the superresolution approach brings low-resolution images closer to high-resolution images rather than completely reproducing them. The superresolution method allows for good

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.

recovery of the uniform area of the image. However, due to the sudden change in the characteristics in the marginal zone, this arrangement poses some difficulties for image restoration. The image restored by the super-resolution algorithm will be blurry for the edge portion, which will cause the features to be blurred. Therefore, the division of the boundary categories is also confusing.

This study compared the CNLUCC 2010 data with the super-resolution method and the traditional land extraction classification map extracted from the Beijing–Tianjin–Hebei region. We could determine that the superresolution method in spatial expression is consistent with the classification result extracted by the traditional method. In addition, the spatial distribution details are also expressed well. These two results are also roughly the same as the CNLUCC data results. Therefore, the super-resolution method is a feasible way to scale conversion and LC classification from 5 to 1 km.

E. Focal Loss Weight Selection

The focal loss can improve the sample imbalance by changing the training weights. However, for multicategory classifications, there is currently no uniform standard for the combination of weights (alpha) for different categories in focal loss [82]–[84]. In general, it is selected based on the experience of the researcher. This approach has some contingencies. Researchers could miss the best combination.

This study explores the optimal combination of weights for different categories under focal loss. It is assumed that the weights are combined into optimal weight combinations according to the percentage of sample percentages, and a set of experiments are designed. We first set up several sets of weight combinations based on the difference in the number of samples between different categories. The main basis is that the smaller number of categories has a greater weight. We use these combinations as a comparison test. The training samples are then counted, and the percentage of each category relative to the total number of samples is calculated. The reciprocal of the percentage is assigned to each category as the weight of the category. This value constitutes a percentage ratio weight combination. A model is then constructed separately for training. We calculated the classification accuracy of the models under different weight combinations and expressed the accuracy of each category in each model and then expressed it through a statistical graph. As shown in Fig. 10, the weighting combination based on the percentage of sample percentages is the best in each category. From this finding, we can determine that when performing multicategory classification, we can use this method to assign the alpha parameter of focal loss to obtain the best results.

We calculated the standard deviation in the accuracy of the category obtained for the different weight combinations in each category. They were 0.002, 0.011, 0.014, 0.009, 0.028, 0.031, 0.042, and 0.051, respectively. The standard deviation of the four categories (cropland, forest, shrubs, and grassland) with a larger sample size was 0.009, and the standard deviation of the four categories (water, wetland, impervious, and bare land) with a small sample size was 0.038. In the comparative

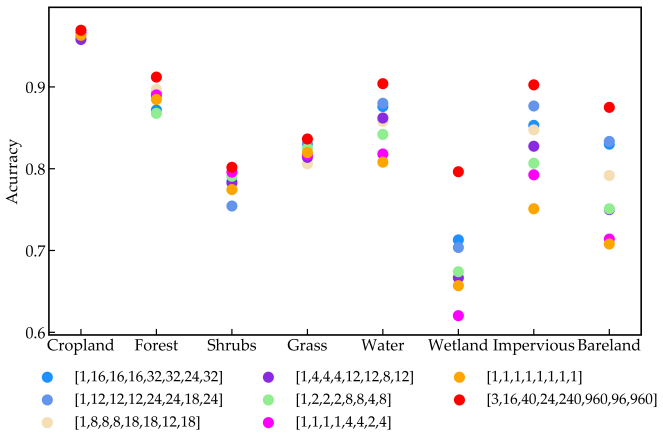


Fig. 10. Influence of the focus loss under different weight combinations on the test accuracy of the small sample categories (water, wetland, impervious, and bare land). For categories with a sufficient sample size (cropland, forest, shrubs, grass), the effect of the weight changes on the accuracy is not obvious. For categories with a small sample size (water, wetland, impervious, and bare land), increasing the weight can improve the accuracy. The Y-axis of the histogram is the classification accuracy, and the X-axis is the category of the verification point.

experiment, the weight of the category with smaller sample size gradually increased. Through the results, we found that the accuracy of these categories basically increases as the weight increases. According to the standard deviation statistics, the accuracy improvement in these categories is larger and more obvious than the category with more sample sizes. Simultaneously, in experiments with weights set according to the percentage of samples, the difference in accuracy between different LC types is minimal. This result provided the best balance in terms of accuracy in different types. The primary reason may be that this combination of weights can make the model's training attention more balanced for each category, thereby yielding a balanced training effect. Therefore, focal loss has little effect on the accuracy of the category with large sample size. For the categories with smaller sample sizes, the recognition accuracy of the model can be improved.

F. Feature Space Visualization

Deep learning is a black box process [85]. It is difficult to know how the model is trained. It is also difficult to know how the model processes and divides the data. However, there are many ways to visualize the output of a model or a process, which can help us monitor the model training process [86].

When we want to classify high-dimensional data, it is not clear whether the dataset has good separability (the distance between similar types is small, and the distance between different types is large) [87]. The separability can be visualized by projecting t-SNE [88] into a 2-D or 3-D space. The primary advantage of the t-SNE is that it can guarantee a similar spatial distribution of points after transformation from high-dimensional space to low-dimensional space. It is a nonlinear algorithm that is very suitable for dimensionality reduction.

To visualize the classification ability of different models, we input the quantitative remote sensing features that were converted from the trained model into the t-SNE and generate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.

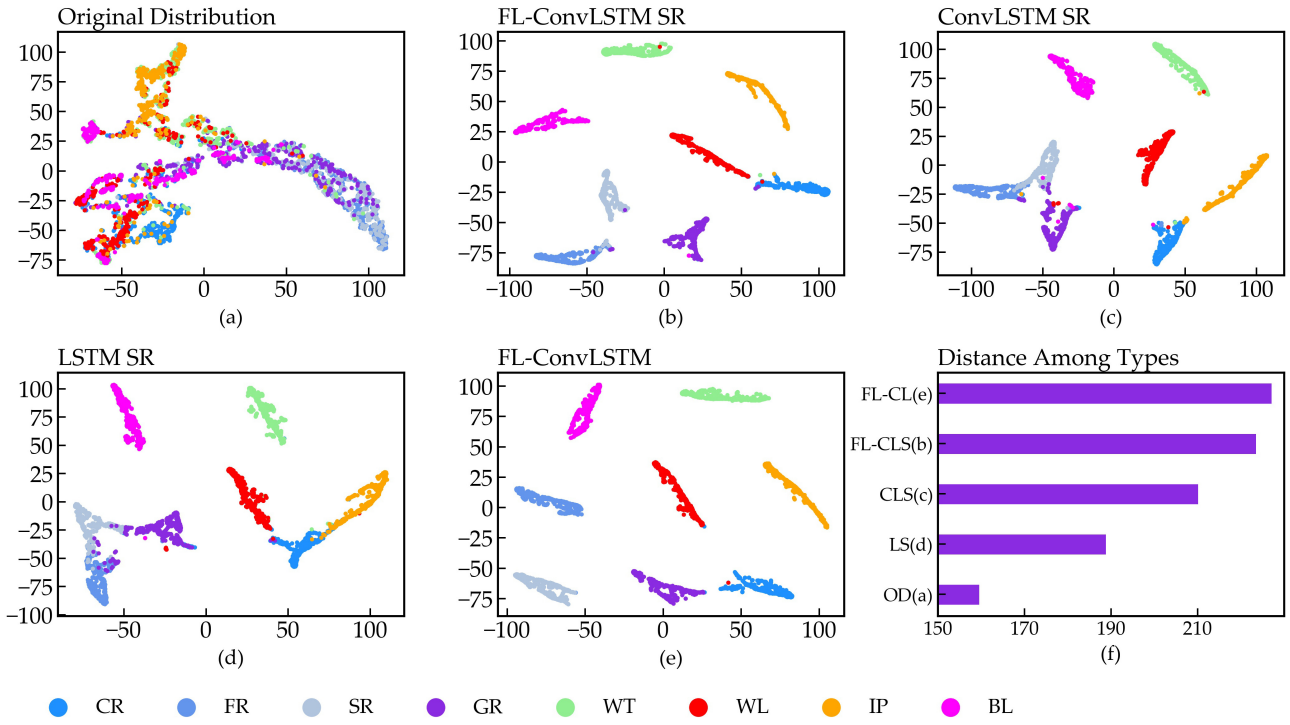


Fig. 11. (a) Feature spatial distribution of the original sample features and sample features after different model transformations. Sample original feature distribution (OD). (b) Feature distribution transformed by the FL-T-ConvLSTM SR model (FL-CLS). (c) Feature distribution transformed by the T-ConvLSTM SR model (CLS). (d) Feature distribution transformed by the LSTM SR model (LS). (e) Feature distribution transformed by the FL-T-ConvLSTM model (FL-CL). (f) Histogram of feature class distances for different models. The Y-axis of the histogram is the classification accuracy, and the X-axis is the category of the verification point.

the spatial distribution of the transformed characteristic parameters of different models (Fig. 11). For the original quantitative remote sensing parameters, the quantitative remote sensing features of different categories have a certain degree of dispersion [89] without model conversion. However, overall, it is still more difficult to distinguish among them. After the transformation of the model, the characteristic parameters of different categories have a good degree of aggregation and differentiation. Among them, water bodies and bare land have good distinguishability in each model, which primarily depends on the surface characteristics of these two categories being different from the other categories.

For different models, the LSTM model distinguishes well between water and bare ground. However, due to the similarity in the surface features, the three categories of grassland, shrub, and forest make LSTM less distinguishable. In addition, LSTM is not excellent at the classification performance of the three categories because of the cross-distribution of the impervious land and the wetland at the junction with the cropland in the study area. These phenomena have been significantly improved after using the T-ConvLSTM model. This finding proves that T-ConvLSTM is indeed an effective method for classification after extracting feature details. To verify this point, the average distance between the classes of the feature distribution map of the t-SNE of different models is counted. From this finding, it is clear that the class spacing of T-ConvLSTM is significantly better than that of LSTM. In addition, we also compare the average distance between the classes of the FL-T-ConvLSTM feature distribution map of t-SNE under super-resolution and traditional methods. Both are well-differentiated for each category and have roughly the same performance. This finding further validates the view that superresolution LC classification can achieve results comparable to those of the traditional methods under the scale conversion from 0.05° to 1 km.

G. Uncertainty and Prospects for Sample Selection

In many classification processes, inappropriate training samples are indeed considered to be the primary source of errors [90]. Changing the number of layers in a neural network has a far smaller impact on the model accuracy than a very efficient set of training samples [91]. Therefore, the selection of training samples is very important. To ensure the reliability of the samples, this study ensures the reliability and accuracy of the samples by extracting the unchanged pixels in the CNLUCC dataset during different years. However, due to the uncertainty of the existing classification datasets, selecting samples directly still cannot fully guarantee the accuracy of the samples [92], [93]. Therefore, improving the sample selection method will be helpful in improving the accuracy and quality of the LC classification [94].

This study presents a new idea for sample selection from the dimension reduction method. After the quantitative remote sensing parameters were converted by FL-T-ConvLSTM, each category was well-distinguished. However, in these clusters

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.

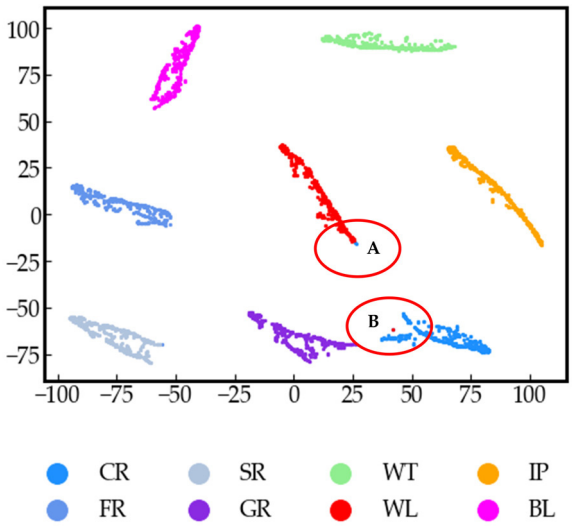


Fig. 12. FL-T-ConvLSTM model dimensionality reduction scatters. Red circle A represents a sample of cropland that is misclassified as wetland. Red circle B represents a sample of wetland that is misclassified as cropland. The vertical and horizontal coordinates represent the relative distance between scattered points.

of categories, there will always be a few other categories of scattered instances that are distributed within that category. Since the quantitative features have been transformed by the model, there is good separability between the classes. In the feature space, they represent the aggregation within the class and the separation between the classes. In addition, there is basically no inter-class connection or adjacent phenomenon, and the expression in the feature space is such that the distances between the classes are small. Therefore, the occurrence of the misclassification of these few scattered instances can rule out the similarity in features. The more likely possibility is that the category label of the feature is incorrect (Fig. 12). After the model is learned and converted, the characteristics of the point are still classified into other categories. This finding indicates that the feature of the point belongs to the inside of the cluster in which it is located, but the label of the point has an error. Thus, the label is the wrong sample. Therefore, this study believes that sample filtering by feature dimensionality reduction clustering of feature vectors is a feasible new method.

V. Conclusion

This study proposes an FL-T-ConvLSTM-based super-resolution long-term series LC model. First, a set of LC classification samples based on invariant pixels were screened out. Based on this action, the correspondence between the 0.05° quantitative remote sensing products and the 1-km LC type was established. Afterward, we trained the super-resolution model and evaluated its robustness.

To make full use of the time series information, this study improved the structure based on LSTM. By combining the LSTM model with the CNN, a set of LSTM models for filtering and feature extraction of time series information were constructed. To solve the sample imbalance problem, the focal loss algorithm was added during the model training process. After the comparison tests, FL-T-ConvLSTM achieved higher accuracy.

In this study, FL-T-ConvLSTM was used as the basic model to train the superresolution method and the traditional method. After the model was stabilized, the accuracy of the two methods reached 90.1% and 91.2%, respectively. This finding shows that the superresolution method can not only have the same precision as the traditional method but also obtain very good results at the scale of 0.05° to 1 km. Simultaneously, for the ten categories to be classified here, the accuracy of five categories reached more than 80%. Among them, the forest and cropland categories reached 90% accuracy. To ensure accuracy, we also evaluated the accuracy of the same category during different years. The error for the ten categories does not exceed 5.1%, and the overall accuracy error does not exceed 0.7%. This study proves the stability of the model for use in time series. To demonstrate the true reliability of the superresolution method, this study also screened a set of field validation samples from Google Earth. The validation accuracy of these samples is 86.78%, indicating that the FL-T-ConvLSTM-based superresolution LC classification is effective.

Acknowledgment

The authors greatly appreciate the Data Center for Resources and Environmental Sciences, Chinese Academy of Sciences (RESDC) for providing the land use and land cover data. They also appreciate the "National Earth System Science Data Center, National Science & Technology Infrastructure of China (http://www.geodata.cn)" for the data support.

References

[1] S. W. Running, "Ecosystem disturbance, carbon, and climate," *Science*, vol. 321, no. 5889, pp. 652–653, 2008.
[2] S. M. Sterling, A. Ducharne, and J. Polcher, "The impact of global land-cover change on the terrestrial water cycle," *Nature Climate Change*, vol. 3, no. 4, pp. 385–390, Apr. 2013.
[3] J. J. Feddema, "The importance of land-cover change in simulating future climates," *Science*, vol. 310, no. 5754, pp. 1674–1678, Dec. 2005.
[4] S. Jin, L. Yang, Z. Zhu, and C. Homer, "A land cover change detection and classification protocol for updating Alaska NLCD 2001 to 2011," *Remote Sens. Environ.*, vol. 195, pp. 44–55, Jun. 2017.
[5] M. A. Wulder *et al.*, "Landsat continuity: Issues and opportunities for land cover monitoring," *Remote Sens. Environ.*, vol. 112, no. 3, pp. 955–969, Mar. 2008.
[6] Z. Zhu and C. E. Woodcock, "Continuous change detection and classification of land cover using all available Landsat data," *Remote Sens. Environ.*, vol. 144, pp. 152–171, Mar. 2014.
[7] M. C. Hansen, R. S. Defries, J. R. G. Townshend, and R. Sohlberg, "Global land cover classification at 1 km spatial resolution using a classification tree approach," *Int. J. Remote Sens.*, vol. 21, nos. 6–7, pp. 1331–1364, Jan. 2000.
[8] M. Herold, P. Mayaux, C. E. Woodcock, A. Baccini, and C. Schmullius, "Some challenges in global land cover mapping: An assessment of agreement and accuracy in existing 1 km datasets," *Remote Sens. Environ.*, vol. 112, no. 5, pp. 2538–2556, May 2008.
[9] H. D. Matthews, A. J. Weaver, K. J. Meissner, N. P. Gillett, and M. Eby, "Natural and anthropogenic climate change: Incorporating historical land cover change, vegetation dynamics and the global carbon cycle," *Climate Dyn.*, vol. 22, no. 5, pp. 461–479, May 2004.
[10] B. Reyers, P. J. O'Farrell, R. M. Cowling, B. N. Egoh, D. C. Le Maitre, and J. H. J. Vlok, "Ecosystem services, land-cover change, and stakeholders: Finding a sustainable foothold for a semiarid biodiversity hotspot," *Ecology Soc.*, vol. 14, no. 1, pp. 38–60, 2009.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.

[11] A. K. Skidmore *et al.*, "Environmental science: Agree on biodiversity metrics to track from space," *Nature*, vol. 523, no. 7561, pp. 403–405, Jul. 2015.

[12] Z. Zhu *et al.*, "Understanding an urbanizing planet: Strategic directions for remote sensing," *Remote Sens. Environ.*, vol. 228, pp. 164–182, Jul. 2019.

[13] P. Gong *et al.*, "Finer resolution observation and monitoring of global land cover: First mapping results with Landsat TM and ETM+ data," *Int. J. Remote Sens.*, vol. 34, no. 7, pp. 2607–2654, 2013.

[14] P. Gong *et al.*, "Stable classification with limited sample: Transferring a 30-m resolution sample set collected in 2015 to mapping 10-m resolution global land cover in 2017," *Sci. Bull.*, vol. 64, no. 6, pp. 370–373, Mar. 2019.

[15] X. Zhang, L. Liu, Y. Wang, Y. Hu, and B. Zhang, "A SPECLib-based operational classification approach: A preliminary test on China land cover mapping at 30 m," *Int. J. Appl. Earth Obs. Geoinf.*, vol. 71, pp. 83–94, 2018.

[16] J. Chen, X. Cao, S. Peng, and H. Ren, "Analysis and applications of GlobeLand30: A review," *ISPRS Int. Geo-Inf.*, vol. 6, no. 8, pp. 230–246, 2017.

[17] M. A. Friedl *et al.*, "MODIS collection 5 global land cover: Algorithm refinements and characterization of new datasets," *Remote Sens. Environ.*, vol. 114, no. 1, pp. 168–182, Jan. 2010.

[18] W. Li, P. Ciais, N. MacBean, S. Peng, P. Defourny, and S. Bontemps, "Major forest changes and land cover transitions based on plant functional types derived from the ESA CCI land cover product," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 47, pp. 30–39, May 2016.

[19] X. Xu, J. Liu, S. Zhang, R. Li, C. Yan, S. Wu, "Remote sensing monitoring data set for land use and cover in China," in *Proc. Data Registration Publishing Syst. Resource Environ. Sci. Data Center Chin. Acad. Sci.*, 2018. [Online]. Available: http://www.resdc.cn, doi: 10.12078/2018070201.

[20] P. M. Smith, "The NOAA/NASA pathfinder AVHRR 8-km land dataset," *Photogram. Eng. Remote Sens.*, vol. 63, pp. 12–32, 1997.

[21] Y. He, E. Lee, and T. A. Warner, "A time series of annual land use and land cover maps of China from 1982 to 2013 generated using AVHRR GIMMS NDVI3g data," *Remote Sens. Environ.*, vol. 199, pp. 201–217, Sep. 2017.

[22] H. Liu, P. Gong, J. Wang, N. Clinton, Y. Bai, and S. Liang, "Annual dynamics of global land cover and its long-term changes from 1982 to 2015," *Earth Syst. Sci. Data*, vol. 12, no. 2, pp. 1217–1243, Jun. 2020.

[23] X. Zhang, L. Liu, X. Chen, S. Xie, and Y. Gao, "Fine land-cover mapping in China using Landsat datacube and an operational SPECLib-based approach," *Remote Sens.*, vol. 11, no. 9, p. 1056, May 2019.

[24] R. Jong, J. Verbesselt, M. E. Schaepman, and S. Bruin, "Trend changes in global greening and browning: Contribution of short-term trends to longer-term change," *Global Change Biol.*, vol. 18, no. 2, pp. 642–655, Feb. 2012.

[25] R. Prestele *et al.*, "Hotspots of uncertainty in land-use and land-cover change projections: A global-scale model comparison," *Glob. Change Biol.*, vol. 22, no. 12, pp. 3967–3983, 2016.

[26] J. Rogan, J. Franklin, and D. A. Roberts, "A comparison of methods for monitoring multitemporal vegetation change using thematic mapper imagery," *Remote Sens. Environ.*, vol. 80, no. 1, pp. 143–156, Apr. 2002.

[27] Z. Zhu *et al.*, "Including land cover change in analysis of greenness trends using all available Landsat 5, 7, and 8 images: A case study from Guangzhou, China (2000–2014)," *Remote Sens. Environ.*, vol. 185, pp. 243–257, Nov. 2016.

[28] D. Wu *et al.*, "Time-lag effects of global vegetation responses to climate change," *Global Change Biol.*, vol. 21, no. 9, pp. 3520–3531, Sep. 2015.

[29] E. F. Lambin *et al.*, "The causes of land-use and land-cover change: Moving beyond the myths," *Global. Environ. Chang.*, vol. 11, no. 4, pp. 261–269, 2001.

[30] N. Ramankutty *et al.*, "Global land-cover change: Recent progress, remaining challenges," in *Land-Use and Land-Cover Change*. Berlin, Germany: Springer, 2006, pp. 9–39.

[31] C. Giri, B. Pengra, J. Long, and T. R. Loveland, "Next generation of global land cover characterization, mapping, and monitoring," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 25, pp. 30–37, Dec. 2013.

[32] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2016.

[33] Y. Hu *et al.*, "An improved algorithm for imbalanced data and small sample size classification," *J. Data Anal. Inf. Process.*, vol. 3, no. 3, pp. 27–33, 2015.

[34] Y. Jia, Y. Ge, Y. Chen, S. Li, G. B. M. Heuvelink, and F. Ling, "Super-resolution land cover mapping based on the convolutional neural network," *Remote Sens.*, vol. 11, no. 15, p. 1815, Aug. 2019.

[35] X. Li, F. Ling, G. M. Foody, Y. Ge, Y. Zhang, and Y. Du, "Generating a series of fine spatial and temporal resolution land cover maps by fusing coarse spatial resolution remotely sensed images and fine spatial resolution land cover maps," *Remote Sens. Environ.*, vol. 196, pp. 293–311, Jul. 2017.

[36] F. Ling and G. M. Foody, "Super-resolution land cover mapping by deep learning," *Remote Sens. Lett.*, vol. 10, no. 6, pp. 598–606, Jun. 2019.

[37] Z. Shao, J. Cai, P. Fu, L. Hu, and T. Liu, "Deep learning-based fusion of Landsat-8 and Sentinel-2 images for a harmonized surface reflectance product," *Remote Sens. Environ.*, vol. 235, Dec. 2019, Art. no. 111425.

[38] X. Li *et al.*, "Optimal endmember-based super-resolution land cover mapping," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 8, pp. 1279–1283, Aug. 2019.

[39] F. Ling, Y. Du, X. Li, W. Li, F. Xiao, and Y. Zhang, "Interpolation-based super-resolution land cover mapping," *Remote Sens. Lett.*, vol. 4, no. 7, pp. 629–638, Jul. 2013.

[40] C. M. Ward, J. Harguess, B. Crabb, and S. Parameswaran, "Image quality assessment for determining efficacy and limitations of super-resolution convolutional neural network (SRCNN)," in *Proc. Appl. Digit. Image Process. XL*, vol. 103960, May 2017.

[41] W. Shi *et al.*, "Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network," in *Proc. CVPR*, 2016, pp. 1874–1883.

[42] J. Gu, X. Sun, Y. Zhang, K. Fu, and L. Wang, "Deep residual squeeze and excitation network for remote sensing image super-resolution," *Remote Sens.*, vol. 11, no. 15, p. 1817, Aug. 2019.

[43] S. Liang *et al.*, "A long-term Global LAnd Surface Satellite (GLASS) data-set for environmental studies," *Int. J. Digit. Earth*, vol. 6, no. sup1, pp. 5–33, Dec. 2013.

[44] L. Zhong, L. Hu, and H. Zhou, "Deep learning based multi-temporal crop classification," *Remote Sens. Environ.*, vol. 221, pp. 430–443, Feb. 2019.

[45] C. Gómez, J. C. White, and M. A. Wulder, "Optical remotely sensed time series data for land cover classification: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 116, pp. 55–72, Jun. 2016.

[46] A. Graves and J. Schmidhuber, "Framewise phoneme classification with bidirectional LSTM and other neural network architectures," *Neural Netw.*, vol. 18, nos. 5–6, pp. 602–610, Jul. 2005.

[47] M. Rußwurm and M. Körner, "Multi-temporal land cover classification with long short-term memory neural networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42W1, p. 551, May 2017.

[48] H. Wang, X. Zhao, X. Zhang, D. Wu, and X. Du, "Long time series land cover classification in China from 1982 to 2015 based on Bi-LSTM deep learning," *Remote. Sens.*, vol. 11, pp. 1639–1660, Jan. 2019.

[49] S. Yu, S. Jia, and C. Xu, "Deep convolutional neural networks for hyperspectral image classification," *Neurocomputing*, vol. 219, pp. 88–98, May 2017.

[50] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote. Sens.*, vol. 54, no. 10, pp. 6232–6251, Jul. 2016.

[51] G. Scarpa, M. Gargiulo, A. Mazza, and R. Gaetano, "A CNN-based fusion method for feature extraction from sentinel data," *Remote Sens.*, vol. 10, no. 2, p. 236, Feb. 2018.

[52] G. Cheng, Z. Li, J. Han, X. Yao, and L. Guo, "Exploring hierarchical convolutional features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6712–6722, Nov. 2018.

[53] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.

[54] X. Shi, Z. Chen, H. Wang, D. Yeung, W. Wong, and W. Woo, "Convolutional LSTM Network: A machine learning approach for precipitation nowcasting," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 802–810.

[55] D. Ienco, R. Gaetano, C. Dupaquier, and P. Maurel, "Land cover classification via multitemporal spatial data by deep recurrent neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1685–1689, Oct. 2017.

[56] F. Karim, S. Majumdar, H. Darabi, and S. Chen, "LSTM fully convolutional networks for time series classification," *IEEE Access*, vol. 6, pp. 1662–1669, 2018.

[57] T. Wiatowski and H. Bölcskei, "A mathematical theory of deep convolutional neural networks for feature extraction," *IEEE Trans. Inf. Theory*, vol. 64, no. 3, pp. 1845–1866, Mar. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.

[58] W. Lee, C.-H. Jun, and J.-S. Lee, "Instance categorization by support vector machines to adjust weights in AdaBoost for imbalanced data classification," *Inf. Sci.*, vol. 381, pp. 92–103, Mar. 2017.
[59] Z. Sun, L. Di, and H. Fang, "Using long short-term memory recurrent neural network in land cover classification on Landsat and cropland data layer time series," *Int. J. Remote Sens.*, vol. 40, no. 2, pp. 593–614, Jan. 2019.
[60] W.-C. Lin, C.-F. Tsai, Y.-H. Hu, and J.-S. Jhang, "Clustering-based undersampling in class-imbalanced data," *Inf. Sci.*, vols. 409–410, pp. 17–26, Oct. 2017.
[61] J. A. Sáez, B. Krawczyk, and M. Woźniak, "Analyzing the oversampling of different classes and types of examples in multi-class imbalanced datasets," *Pattern Recognit.*, vol. 57, pp. 164–178, Sep. 2016.
[62] R. Li, X. Xiao, S. Ni, H. Zheng, and S. Xia, "Byte segment neural network for network traffic classification," in *Proc. IEEE/ACM 26th Int. Symp. Qual. Service (IWQoS)*, Banff, AB, Canada, 2018, pp. 1–10.
[63] A. H. Strahler *et al.*, "Global land cover validation: Recommendations for evaluation and accuracy assessment of global land cover maps," *Eur. Communities, Luxembourg*, vol. 51, no. 4, pp. 1–15, 2006.
[64] P. Olofsson *et al.*, "A global land-cover validation data set, part I: Fundamental design principles," *Int. J. Remote Sens.*, vol. 33, no. 18, pp. 5768–5788, Sep. 2012.
[65] Y. Zhao *et al.*, "Towards a common validation sample set for global land-cover mapping," *Int. J. Remote Sens.*, vol. 35, no. 13, pp. 4795–4814, Jul. 2014.
[66] R. K. Panigrahy, M. P. Kale, U. Dutta, A. Mishra, B. Banerjee, and S. Singh, "Forest cover change detection of Western Ghats of Maharashtra using satellite remote sensing based visual interpretation technique," *Current Sci.*, vol. 98, no. 5, pp. 657–664, 2010.
[67] A. Shalaby and R. Tateishi, "Remote sensing and GIS for mapping and monitoring land cover and land-use changes in the northwestern coastal zone of Egypt," *Appl. Geography*, vol. 27, no. 1, pp. 28–41, Jan. 2007.
[68] A. Shukla, M. K. Arora, and R. P. Gupta, "Synergistic approach for mapping debris-covered glaciers using optical–thermal remote sensing data with inputs from geomorphometric parameters," *Remote Sens. Environ.*, vol. 114, no. 7, pp. 1378–1387, Jul. 2010.
[69] Z. Xiao, S. Liang, J. Wang, Y. Xiang, X. Zhao, and J. Song, "Long-time-series global land surface satellite leaf area index product derived from MODIS and AVHRR surface reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 9, pp. 5301–5318, Sep. 2016.
[70] K. Nishida, R. R. Nemani, J. M. Glassy, and S. W. Running, "Development of an evapotranspiration index from Aqua/MODIS for monitoring surface moisture status," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 2, pp. 493–501, Feb. 2003.
[71] Z. Xiao, S. Liang, R. Sun, J. Wang, and B. Jiang, "Estimating the fraction of absorbed photosynthetically active radiation from the MODIS data based GLASS leaf area index product," *Remote Sens. Environ.*, vol. 171, pp. 105–117, Dec. 2015.
[72] T. He, S. Liang, and D.-X. Song, "Analysis of global land surface albedo climatology and spatial-temporal variation during 1981–2010 from multiple satellite products," *J. Geophys. Res. Atmos.*, vol. 119, no. 17, pp. 10281–10298, Sep. 2014.
[73] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.
[74] P. O. Gislason, J. A. Benediktsson, and J. R. Sveinsson, "Random forests for land cover classification," *Pattern Recognit. Lett.*, vol. 27, no. 4, pp. 294–300, Mar. 2006.
[75] C. Huang, L. S. Davis, and J. R. G. Townshend, "An assessment of support vector machines for land cover classification," *Int. J. Remote Sens.*, vol. 23, no. 4, pp. 725–749, Jan. 2002.
[76] K. Hayat, "Multimedia super-resolution via deep learning: A survey," *Digit. Signal Process.*, vol. 81, pp. 198–217, Oct. 2018.
[77] N. D. Le and J. V. Zidek, "Interpolation with uncertain spatial covariances: A Bayesian alternative to kriging," *J. Multivariate Anal.*, vol. 43, no. 2, pp. 351–374, Nov. 1992.
[78] R. Congalton, J. Gu, K. Yadav, P. Thenkabail, and M. Ozdogan, "Global land cover mapping: A review and uncertainty analysis," *Remote Sens.*, vol. 6, no. 12, pp. 12070–12093, Dec. 2014.
[79] J. Townshend, C. Justice, W. Li, C. Gurney, and J. Mcmanus, "Global land cover classification by remote sensing: Present capabilities and future possibilities," *Remote Sens. Environ.*, vol. 35, nos. 2–3, pp. 243–255, Feb. 1991.
[80] Y. Shao and R. S. Lunetta, "Sub-pixel mapping of tree canopy, impervious surfaces, and cropland in the laurentian great lakes basin using MODIS time-series data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 4, no. 2, pp. 336–347, Jun. 2011.
[81] G. M. Foody, A. M. Muslim, and P. M. Atkinson, "Super-resolution mapping of the waterline from remotely sensed data," *Int. J. Remote Sens.*, vol. 26, no. 24, pp. 5381–5392, Dec. 2005.
[82] J. Chang *et al.*, "Brain tumor segmentation based on 3D unet with multi-class focal loss," in *Proc. 11th Int. Congr. Image Signal Process., BioMed. Eng. Inform. (CISP-BMEI)*, Beijing, China, 2018, pp. 1–5.
[83] Y. Cui, M. Jia, T. Lin, Y. Song, and S. Belongie, "Class-balanced loss based on effective number of samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, 2019, pp. 9260–9269.
[84] R. Qin, K. Qiao, L. Wang, L. Zeng, J. Chen, and B. Yan, "Weighted focal loss: An effective loss function to overcome unbalance problem of chest X-ray14," in *Proc. IOP Conf. Ser., Mater. Sci. Eng.*, vol. 428, Oct. 2018, pp. 12–22.
[85] D. Michie, D. J. Spiegelhalter, and C. Taylor, "Machine learning," *Neural Stat. Classification*, vol. 13, pp. 1–298, Feb. 1994.
[86] F. Jia, Y. Lei, N. Lu, and S. Xing, "Deep normalized convolutional neural network for imbalanced fault classification of machinery and its understanding via visualization," *Mech. Syst. Signal Process.*, vol. 110, pp. 349–367, Sep. 2018.
[87] P. Zhou, J. Han, G. Cheng, and B. Zhang, "Learning compact and discriminative stacked autoencoder for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4823–4833, Jul. 2019.
[88] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, Jan. 2008.
[89] X. Wu, Q. Xiao, J. Wen, D. You, and A. Hueni, "Advances in quantitative remote sensing product validation: Overview and current status," *Earth-Science Rev.*, vol. 196, Sep. 2019, Art. no. 102875.
[90] M. Pal and P. M. Mather, "Some issues in the classification of DAIS hyperspectral data," *Int. J. Remote Sens.*, vol. 27, no. 14, pp. 2895–2916, Jul. 2006.
[91] G. M. Foody and M. K. Arora, "An evaluation of some factors affecting the accuracy of classification by an artificial neural network," *Int. J. Remote Sens.*, vol. 18, no. 4, pp. 799–810, Mar. 1997.
[92] R. Colditz, "An evaluation of different training sample allocation schemes for discrete and continuous land cover classification using decision tree-based algorithms," *Remote Sens.*, vol. 7, no. 8, pp. 9655–9681, Jul. 2015.
[93] Y. Shao and R. S. Lunetta, "Comparison of support vector machine, neural network, and CART algorithms for the land-cover classification using limited training data points," *ISPRS J. Photogramm. Remote Sens.*, vol. 70, pp. 78–87, Jun. 2012.
[94] J. Radoux, C. Lamarche, E. Van Bogaert, S. Bontemps, C. Brockmann, and P. Defourny, "Automated training sample extraction for global land cover mapping," *Remote Sens.*, vol. 6, no. 5, pp. 3965–3987, May 2014.



Haoyu Wang received the bachelor's degree in geomatics engineering from the China University of Mining and Technology, Beijing, China, in 2015, and the master's degree in cartography and geographical information system from Beijing Normal University, Beijing, in 2020. He is pursuing the Ph.D. degree in cartography and geographical information system with Peking University, Beijing.

He has recently worked in long time series land cover classification with multiple resolutions. His main research interest includes land cover classification and deep learning applications of remote sensing from long series of satellite images.



Xiang Zhao received the Ph.D. degree in cartography and geographic information systems from Beijing Normal University, Beijing, China, in 2006.

From 2008 to 2010, he was a Post-Doctoral Fellow with the College of Resources Science and Technology, Beijing Normal University. His main research interests focus on high-performance computing system construction and quantitative remote sensing application. He also did some research on long time series remote sensing data trend analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.



Shunlin Liang (Fellow, IEEE) received the Ph.D. degree from Boston University, Boston, MA, USA.

He is a Professor with the Department of Geographical Sciences, University of Maryland, College Park, MD, USA. He has led a team to develop the Global LAns Surface Satellite (GLASS) products that are publically available and widely used. He has authored more than 370 SCI indexed peer-reviewed journal articles, 42 book chapters, and nine special issues of different journals. He has authored or edited seven books and four of which were translated in Chinese, such as *Quantitative Remote Sensing of Land Surfaces* (Wiley, 2004), *Advances in Land Remote Sensing: System, Modeling, Inversion and Application* (Springer, 2008), *Advanced Remote Sensing: Terrestrial Information Extraction and Applications* (Academic Press, 2012, 2019), *Global LAnd Surface Satellite (GLASS) Products: Algorithms, Validation and Analysis* (Springer, 2013), *Land Surface Observation, Modeling, Data Assimilation* (World Scientific, 2013), and *Earth's Energy Budget* (Elsevier, 2017). His main research interests focus on estimating land surface variables from satellite data, global satellite product generation, Earth's energy balance, and environmental changes.

Dr. Liang was the Editor-in-Chief of the nine-volume books entitled *Comprehensive Remote Sensing* (Elsevier, 2017). He was an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He is an Editor-in-Chief of the *Science of Remote Sensing*.



Donghai Wu received the Ph.D. degree from the College of Urban and Environmental Sciences, Sino-French Institute for Earth System Science, Peking University, Beijing, China, in 2019.

His is a Post-Doctoral Associate with the Department of Ecology and Evolutionary Biology, Cornell University, Ithaca, NY, USA. His research interests include carbon cycle and climate change.



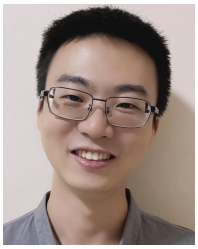
Xin Zhang received the Ph.D. degree in cartography and geographic information system from Beijing Normal University (BNU), Beijing, China, in 2014.

He is an Associate Researcher with The Aerospace Information Research Institute (AIR), Chinese Academy of Sciences (CAS), Beijing. His research interests include remote sensing image processing and deep learning.



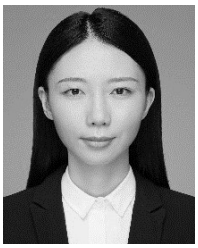
Qian Wang received the B.E. degree in geographic information system from the Beijing University of Civil Engineering and Architecture, Beijing, China, in 2014.

He works with Beijing Normal University, Beijing. His main research interests focus on global quantitative remote sensing product generation on HPC platform.



Jiacheng Zhao received the B.S. degree in electronic information engineering from Zhengzhou University, Zhengzhou, China, in 2014, and the M.S. degree in forest management from the Chinese Academy of Forestry, Beijing, China, in 2018. He is pursuing the Ph.D. degree in cartography and geographical information system with Beijing Normal University, Beijing.

He has recently worked in different biophysical traits of urban trees and their responses to climate and land cover changes. His main research interest includes landscape modeling and urban environment monitoring from long series of satellite images.



Xiaozheng Du received the bachelor's degree in geographical information science from Lanzhou University, Lanzhou, China, in 2017, and the master's degree in cartography and geographical information system from Beijing Normal University, Beijing, China, in 2020.

Her main research interests include the ecological and environmental benefits of restoration projects such as afforestation and monitoring vegetation changes in a long series of satellite images.



Qian Zhou received the B.S. degree from the China University of Mining and Technology-Beijing, Beijing, China, in 2018. She is pursuing the M.S. degree in cartography and geographical information system with Beijing Normal University, Beijing.

Since 2018, she has been a Student with Beijing Normal University. Her research interests include remote sensing, vegetation index, and urbanization, especially the impact of urbanization on vegetation and global vegetation stability.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:38:53 UTC from IEEE Xplore. Restrictions apply.